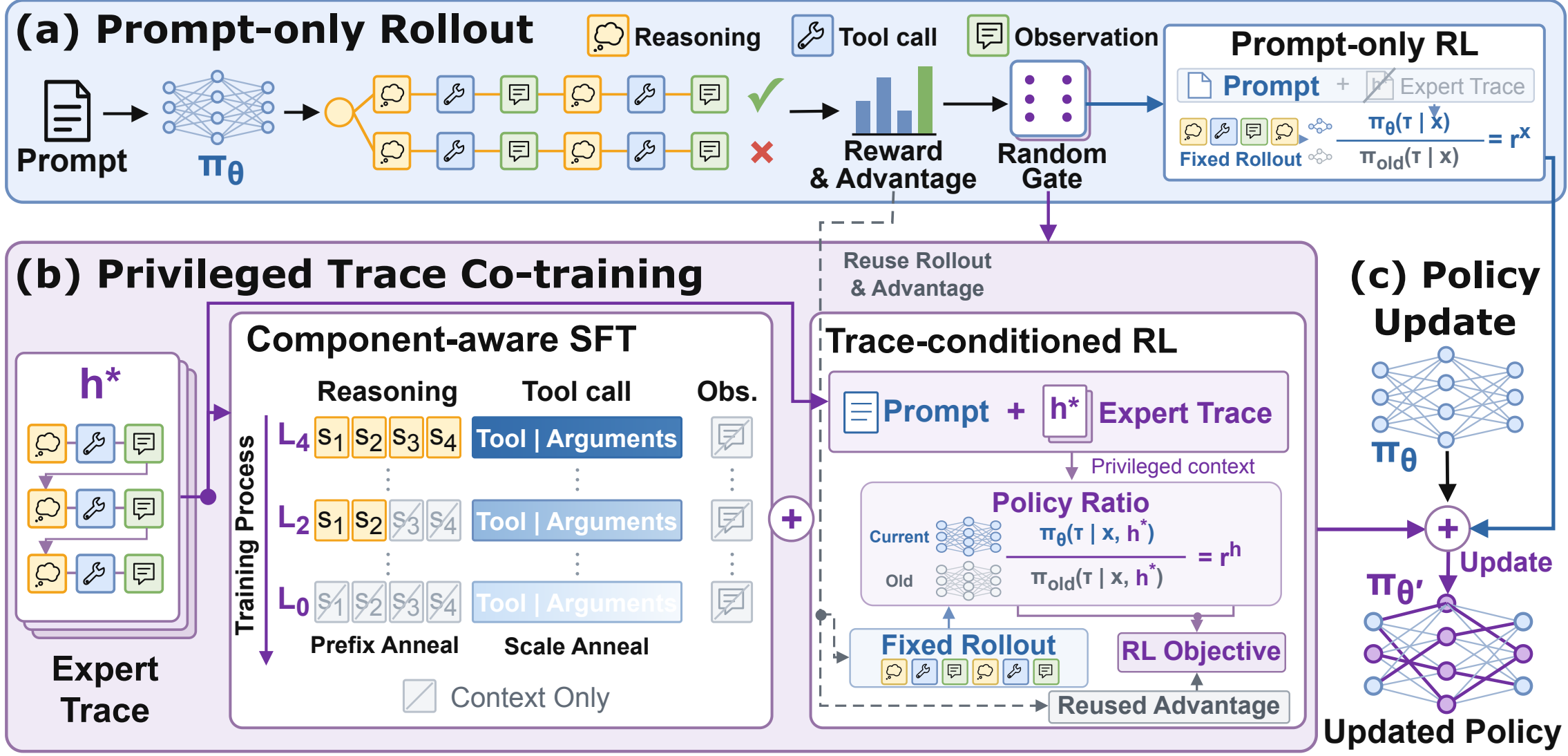

(a) Prompt-only Rollout
Reasoning
Tool call
Observation
Prompt
π_θ
Reward & Advantage
Random Gate
Prompt-only RL
Prompt + Expert Trace
Fixed Rollout
$\frac{\pi_\theta(\tau \mid x)}{\pi_{old}(\tau \mid x)} = r^x$
(b) Privileged Trace Co-training
Reuse Rollout & Advantage
h*
Expert Trace
Component-aware SFT
Reasoning
Tool call
Obs.
Training Process
L_4 s_1 s_2 s_3 s_4 Tool | Arguments
L_2 s_1 s_2 s_3 s_4 Tool | Arguments
L_0 s_1 s_2 s_3 s_4 Tool | Arguments
Prefix Anneal
Scale Anneal
Context Only
Trace-conditioned RL
Prompt + h* Expert Trace
Privileged context
Policy Ratio
Current
Old
$\frac{\pi_\theta(\tau \mid x, h^*)}{\pi_{old}(\tau \mid x, h^*)} = r^h$
Fixed Rollout
RL Objective
Reused Advantage
(c) Policy Update
π_θ
Update
$\pi_{\theta'}$
Updated Policy